

Group members :

1. Afina Soleha Batrisyia bt Mohd Hisafudin

A24CS0034

2. Fakhira Anisa bt Mohd Radzi

A24CS0070

3. Alisya Humayraa bt Mohd Khairani

A24CS0043



CamScanner

Recurrence Relation

1)

$$a_0 = 1$$

$$a_1 = 3(1) + 2 = 5$$

$$a_2 = 3(5) + 2 = 17$$

$$a_3 = 3(17) + 2 = 53$$

$$a_4 = 3(53) + 2 = 161$$

$$a_5 = 3(161) + 2 = 485$$

2)

$$b_1 = 1$$

$$b_2 = 1 + 2^2 = 5$$

$$b_3 = 5 + 3^2 = 14$$

$$b_4 = 14 + 4^2 = 30$$

$$\therefore b_2 = 5, b_3 = 14, b_4 = 30$$

3)

$$a_n^h = 2a_{n-1}^h$$

$$r = 2$$

$$a_n^h = A \cdot 2^n$$

$$a_n^p = c$$

$$c = 2c + 5$$

$$-5 = c$$

$$c = -5$$

$$a_n = A \cdot 2^n - 5$$

$$a_0 = 3$$

$$3 = A \cdot 2^0 - 5$$

$$3 = A - 5$$

$$A = 8$$

$\therefore$ The closed-form solution is $a_n = 8 \cdot 2^n - 5$

4)

$$T_1 = 1$$

$$T_2 = 2(1) + 1 = 3$$

$$T_3 = 2(3) + 1 = 7$$

$$T_4 = 2(7) + 1 = 15$$

$$T_n = 2^{n-1} \text{ for all } n \geq 1$$

$$n = 1$$

$$T_1 = 2^1 - 1$$

$$= 1$$

Assume $n = k$

$$T_k = 2^k - 1$$

$$n = k + 1$$

$$T_{k+1} = 2^{k+1} - 1$$

$$T_{k+1} = 2T_k + 1$$

$$T_{k+1} = 2(2^k - 1) + 1$$

$$= 2 \cdot 2^k - 2 + 1$$

$$= 2^{k+1} - 1$$

$$n = k + 1$$

$\therefore$ The closed-form solution is $T_n = 2^n - 1, n \geq 1$

counting method & probability.

1.

- (a)
- T_1 = choose first digit of code, $n_1 = 10$ ways
 - T_2 = choose second digit of code, $n_2 = 9$ ways
 - T_3 = choose third digit of code, $n_3 = 8$ ways
 - T_4 = choose fourth digit of code, $n_4 = 7$ ways
 - T_5 = choose fifth digit of code, $n_5 = 6$ ways

$$\begin{aligned} T &= n_1 \times n_2 \times n_3 \times n_4 \times n_5 \\ &= 10 \times 9 \times 8 \times 7 \times 6 \\ &= 30240 \text{ ways} \end{aligned}$$

- (b) $P = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$
even number = $\{0, 2, 4, 6, 8\}$

- T_1 = choose first digit even number, $n_1 = 5$ ways
- T_2 = choose second digit of code, $n_2 = 9$ ways
- T_3 = choose third digit of code, $n_3 = 8$ ways
- T_4 = choose fourth digit of code, $n_4 = 7$ ways
- T_5 = choose fifth digit of code, $n_5 = 6$ ways

$$\begin{aligned} T &= n_1 \times n_2 \times n_3 \times n_4 \times n_5 \\ &= 5 \times 9 \times 8 \times 7 \times 6 \\ &= 15120 \text{ ways} \end{aligned}$$

2.
(a) 10 9 8 6 7 5 4 3 2 1

- T_1 = choose first of place, $n_1 = 10$ ways
- T_2 = choose second of place, $n_2 = 9$ ways
- T_3 = choose third of place, $n_3 = 8$ ways
- T_4 = choose fourth of place, $n_4 = 7$ ways
- T_5 = choose fifth of place, $n_5 = 6$ ways
- T_6 = choose six of place, $n_6 = 5$ ways
- T_7 = choose seven of place, $n_7 = 4$ ways
- T_8 = choose eight of place, $n_8 = 3$ ways
- T_9 = choose ninth of place, $n_9 = 2$ ways
- T_{10} = choose ten of place, $n_{10} = 1$ ways

$$\begin{aligned} T &= n_1 \times n_2 \times n_3 \times n_4 \times n_5 \times n_6 \times n_7 \times n_8 \times n_9 \times n_{10} \\ &= 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 \\ &= 3628800 \text{ ways} \end{aligned}$$

(b)

AB	9	8	7	6	5	4	3	2
↑								

two people one place

T_1 : choose two people for one place = $n_1 = 2$ ways

T_2 : choose for second place

$n_2 = 9$ ways

T_3 : choose for third place

$n_3 = 8$ ways

T_4 : choose for fourth place

$n_4 = 7$ ways

T_5 : choose for fifth place

$n_5 = 6$ ways

T_6 : choose for six place

$n_6 = 5$ ways

T_7 : choose for seven place

$n_7 = 4$ ways

T_8 : choose for eight place

$n_8 = 3$ ways

T_9 : choose for ninth place

$n_9 = 2$ ways

$$\begin{aligned}
 T &= n_1 \times n_2 \times n_3 \times n_4 \times n_5 \times n_6 \times n_7 \times n_8 \times n_9 \\
 &= 2 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \\
 &= 72 \times 5760 \text{ ways}
 \end{aligned}$$

2.

two dice $\rightarrow$ six sided

(a) sum of the number is 7.

$$\begin{aligned}
 1 + 6 &= 7 \\
 2 + 5 &= 7 \\
 3 + 4 &= 7 \\
 4 + 3 &= 7 \\
 5 + 2 &= 7 \\
 6 + 1 &= 7
 \end{aligned}$$

$$\begin{aligned}
 \text{two dice} &= 6 \times 6 \\
 &= 36
 \end{aligned}$$

$$1 + 6 = 7$$

$$3 + 4 = 7$$

$$5 + 2 = 7$$

$$6 + 1 = 7$$

$$4 + 3 = 7$$

$$2 + 5 = 7$$

$$\begin{aligned}
 \text{probability of sum of} &= \frac{6}{36} \\
 \text{the number is 7} &= \frac{1}{6}
 \end{aligned}$$

(b) at least one dice shows 6.

$$\text{not number 6} = \{1, 2, 3, 4, 5\}$$

$$\begin{aligned}
 \text{two dice} &= 5 \times 5 \\
 &= 25
 \end{aligned}$$

probability at least one dice

show 6

$$= 36 - 25$$

$$= 11$$

$$= \frac{11}{36}$$



(c) the number of dice are equal.

n, n

$(1, 1)$
 $(2, 2)$
 $(3, 3)$
 $(4, 4)$
 $(5, 5)$
 $(6, 6)$

probability number of
dice are equal $= \frac{6}{36}$

$= \frac{1}{6}$

9.

S T A T I S T I C S

$S = 3$

$T = 3$

$A = 1$

$I = 2$

$C = 1$

(a)

First letter = 'S'

rest of the word = 9

arrangement
starting with 'S' $= \frac{9!}{3! \cdot 2! \cdot 2! \cdot 1! \cdot 1!}$

$= \frac{362880}{24}$

$= 15120$

(a) ${}^nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!}$

$= \frac{15!}{4!(15-4)!}$

$= 1,365 \text{ ways}$

(b)

2 student chosen.

13 student no group.

${}^nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!}$

$\binom{13}{2} = \frac{13!}{2!(13-2)!}$

$= 78 \text{ ways}$

(b)

$T = 3 \rightarrow \underline{T T T}$

arrangement all

'T' together

$= \frac{9!}{3! \cdot 2! \cdot 1! \cdot 1!}$

$= \frac{362880}{12}$

$= 30240$

PERMUTATION & COMBINATION

1. 8 characters from set $\{A, B, C, D, 1, 2, 3, 4, !, @, \#, \$\}$

$$(n) \quad 12 \quad 11 \quad 10 \quad 9 \quad 8 \quad 7 \quad 6 \quad 5$$

$${}^{12}P_8 = \frac{12!}{(12-8)!} = \frac{12!}{4!} = \frac{479001600}{24} = 19958400 \text{ ways}$$

$$(b) \quad 4 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4$$

$$\begin{aligned} \text{no. of passwords} &: 4 \times 4 \times (10 \times 9 \times 8 \times 7 \times 6 \times 5) \\ &= 2419200 \text{ passwords} \end{aligned}$$

2. "ENGINEERING"

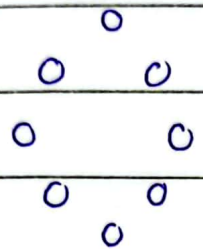
$$\begin{aligned} (a) \quad \frac{11!}{3!2!3!2!} &= \frac{39916800}{144} \\ &= 277200 \text{ ways} \end{aligned}$$

$$(b) \quad \underline{EEE}$$

$$\begin{aligned} \frac{9!}{2!3!2!1!} &= \frac{362880}{24} \\ &= 15120 \text{ ways} \end{aligned}$$

Permutation and Combinations

3.

 8 beads different colour.

a) circular arrangement

$$P(n) = (n-1)!$$

$$P(8) = (8-1)!$$

$$= 7!$$

$$= 5040$$

b) symmetrical arrangement

$$P(n) = \frac{(n-1)!}{2}$$

$$2$$

$$= \frac{(8-1)!}{2}$$

$$2$$

$$= 7!$$

$$2$$

$$= \frac{5040}{2}$$

$$2$$

$$= 2520$$

4. 6 - member of committee
 10 - men
 8 - women

$$\text{Total} = 10 + 8$$

$$n = 18 \text{ people}$$

a) $C(n, r) = {}^nC_r = {}^nC_r$

$$= \binom{n}{r}$$

$$= \frac{n!}{r!(n-r)!}$$

$$= \frac{18!}{6!(18-6)!}$$

$$= \frac{18!}{6! \cdot 12!}$$

$$= 18564$$

$$= 18564$$

$$= 18564$$

$$= 18564$$

b) at least 2 women

- select 2 women $\rightarrow$ exactly 3 women

case 1 : 3 women & 3 men

$$C(8, 4) = \frac{8!}{4!(8-4)!}$$

case 2 : 2 women & 4 men

$$= 70$$

case 1 = 14700 ways

case 2 :

- select 3 men

$$C(8, 2) = \frac{8!}{2!(8-2)!}$$

$$C(10, 4) = \frac{10!}{4!(10-4)!}$$

$$= 28$$

$$= 210$$

$$= 28$$

$$= 210$$

$$C(10, 4) = \frac{10!}{4!(10-4)!}$$

$$\text{case 1} = 210 \cdot 70$$

$$= 210$$

$$= 14700$$

$$= 210$$

$$210 \cdot 28 = 5880$$

$$\text{case 1} + \text{case 2} = 14700 + 5880$$

$$= 20580$$

$$5) a) {}^{30}C_{10} = 30045015$$

$${}^{20}C_{12} = 125970$$

$${}^8C_8 = 1$$

$$30045015 \times 125970 \times 1$$

$$= 3.78 \times 10^{12} \text{ ways}$$

$$b) {}^{29}C_9 = 10015005$$

$${}^{20}C_{12} = 125970$$

$$10015005 \times 125970$$

$$= 1.26 \times 10^{12} \text{ ways}$$

$$6) a) {}^{10}C_5 = 252 \text{ ways}$$

$$b) C(n+r-1, r) = \binom{n+r-1}{r} = \frac{(n+r-1)!}{r!(n-1)!}$$

$$C(14, 5) = \frac{14!}{5!(10-1)!}$$

$$= 2002 \text{ ways}$$

PIGEONHOLE PRINCIPLE

1. Pigeons : set $\{1, 2, 3, \dots, 99\}$ ($n = 99$)

Pigeonholes : 50 integers ($m = 50$)

$$k = \left\lceil \frac{n}{m} \right\rceil = \left\lceil \frac{99}{50} \right\rceil = 2$$

$\therefore$ At least two integers are consecutive [proven]

2. Pigeons : 9 positive integers ($n = 9$)

Pigeonholes : set of remainders $\{0, 1, 2, 3, 4, 5, 6, 7\}$ ($m = 8$)

$$k = \left\lceil \frac{n}{m} \right\rceil = \left\lceil \frac{9}{8} \right\rceil = 2$$

$\therefore$ There are two integers whose difference is divisible by 8 and have the same remainder [proven]

3. Pigeons : 30 students ($n = 30$)

Pigeonholes : days of the week ($m = 7$)

$$k = \left\lceil \frac{n}{m} \right\rceil = \left\lceil \frac{30}{7} \right\rceil = 5$$

$\therefore$ There are 5 students that were born in the same day of the week.

Thus, at least two students were born on the same day of the week.

[proven].

4. Pigeons : 9 socks ($n = 9$)

Pigeonholes : colour of the socks ($m = 3$)

$$k = \left\lceil \frac{n}{m} \right\rceil = \left\lceil \frac{9}{3} \right\rceil = 3$$

$\therefore$ At least 3 socks must be of the same colour.